

2110203 COM ENG MATH II

SIGNAL PROCESSING

OBJECTIVES & CP STUDENT OUTCOMES

2110203 Com Eng Math II (Signals):

Objectives: To be able to understand and apply fundamental signal processing including Fourier transform and analysis

CPSO: (a.2) an ability to apply knowledge of computer mathematics to solve computer engineering problems

2110233 Com Eng Math Lab (Signals):

Objectives: Be able to use Fourier transform for frequency analysis

CPSO: (k.1) an ability to utilise mathematical tools

BOOKS

- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)

1. **Signals & systems** / Alan V. Oppenheim, Alan S. Willsky, with Hamad Nawab.



Book

By: Oppenheim, Alan V.. Upper Saddle River, N.J. : Prentice Hall, c1997. xxx, 957 p. : ill.; 25 cm.
Language: English, Database: CU Library

Subjects: System analysis; Signal theory (Telecommunication)

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Location	Call No.	Status
Engineering Library	QA402 O63 1997 c.3	CHECK SHELVES

[Show More \(2\)](#)

1. **Continuous and discrete signals and systems** / Samir S. Soliman, Mandyam D. Srinath.



Book

By: Soliman, Samir S. Upper Saddle River, NJ : Prentice Hall, 1998 xvii, 525 pages : illustrations ; 25 cm.
Language: English, Database: CU Library

Subjects: Signal theory (Telecommunication); Signal Processing -- Mathematics; System analysis

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[HTTPS://WWW.YOUTUBE.COM/WATCH?V=VVOXAADM0U](https://www.youtube.com/watch?v=VVOXAADM0U)

The video player displays a digital drawing of a square wave in blue and its Fourier series approximation in red. The red curve oscillates around the blue square wave, illustrating the Gibbs phenomenon. The video is titled "Fourier คือเพื่อนเรา" (Fourier is our friend) by the channel "9arm" (1.4M subscribers). The video has 8.2K likes and is shared by several users listed in the sidebar, including Chincoop Cho, Mhaknow, and others. The video progress bar shows 4:20 / 24:18.

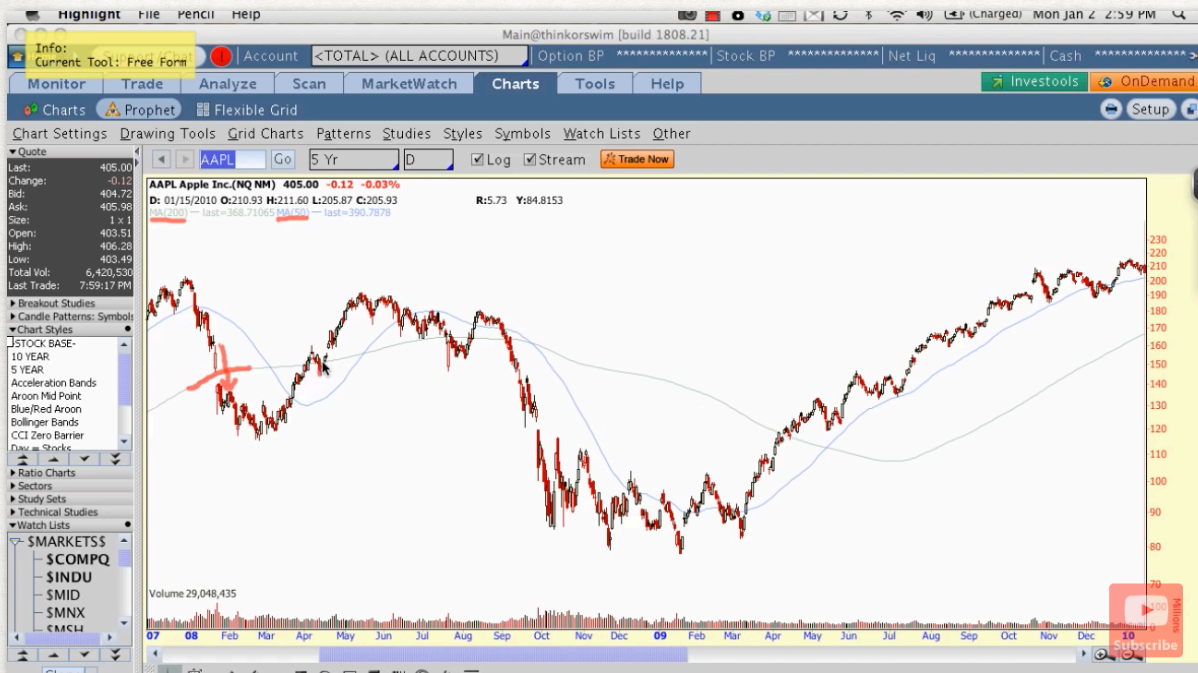
Fourier คือเพื่อนเรา

9arm 1.4M subscribers

Join Subscribe

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SIGNALS EXAMPLES

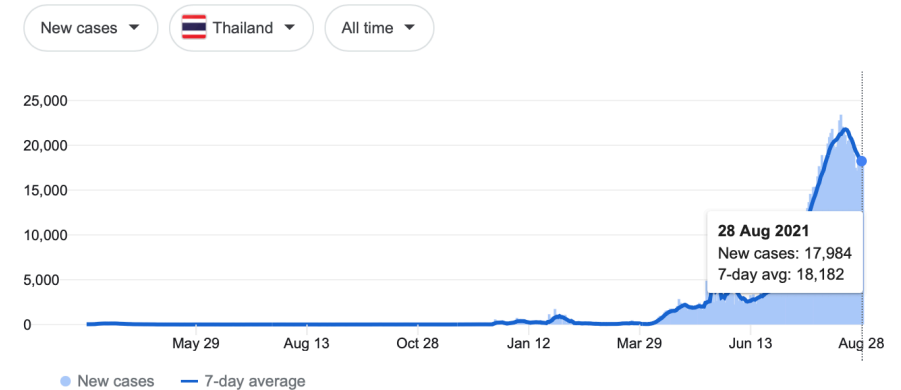


Reference : <https://youtu.be/x4nO5yoarM0>

Statistics

New cases and deaths

From JHU CSSE COVID-19 Data · Last updated: 2 days ago



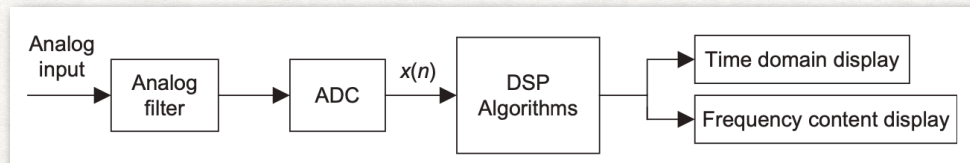
Each day shows new cases reported since the previous day · [About this data](#)

BASIC DIGITAL SIGNAL PROCESSING EXAMPLES

- Digital Filtering



- Signal Frequency (Spectrum) Analysis



Li Tan and Jean Jiang (2019), Digital Signal Processing Fundamentals and Applications (3rd Edition)

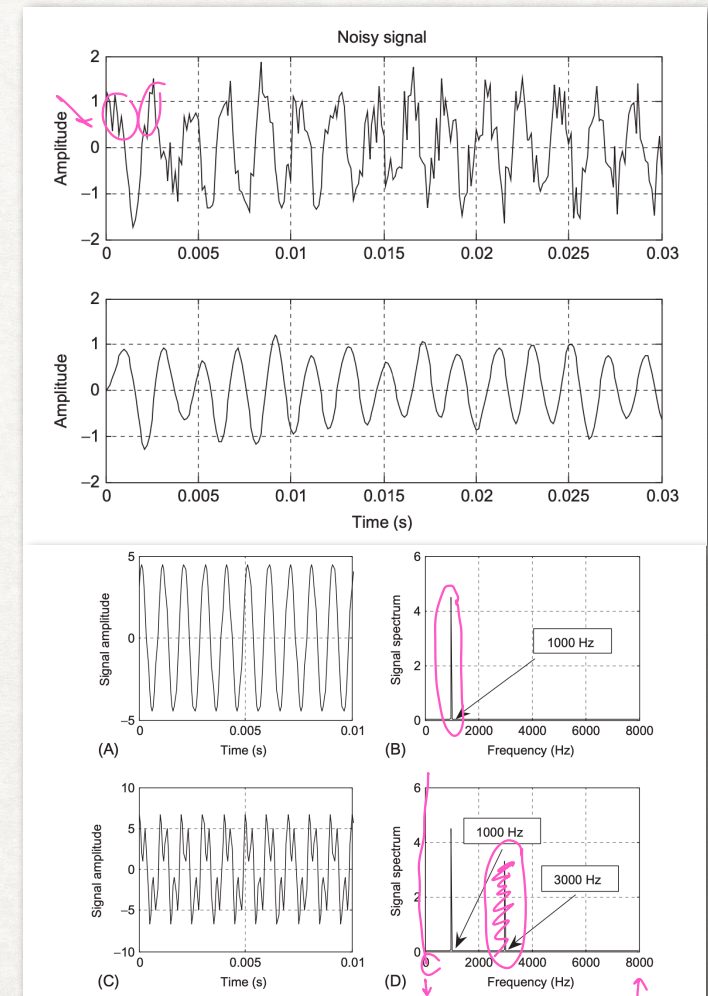


FIG. 1.5
 Audio signals and their spectra. (A) 1000 Hz audio signal. (B) 1000 Hz audio signal spectrum. (C) Audio signal containing 1000 and 3000 Hz frequency components. (D) Audio signal spectrum containing 1000 and 3000 Hz frequency components.

SIGNALS

FREQUENCY DOMAIN

- Noise removal



- <https://youtu.be/keYzcEHugx8>

- Image enhancement

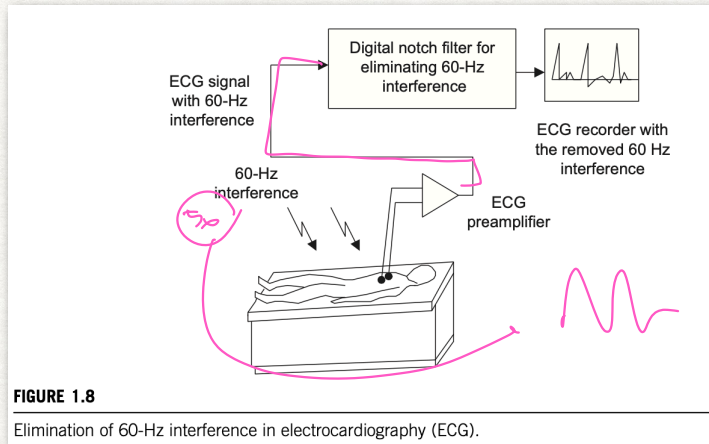


<https://www.osapublishing.org/oe/fulltext.cfm?uri=oe-22-1-618&id=276513>

Thin cloud removal from single satellite images by Jun Liu, et al. (2014)

DIGITAL SIGNAL PROCESSING IN REAL WORLD APPLICATIONS

- Digital Crossover Audio System
- Interference cancellation in electrocardiography



- Speech coding and compression

Li Tan and Jean Jiang (2019), *Digital Signal Processing Fundamentals and Applications (3rd Edition)*

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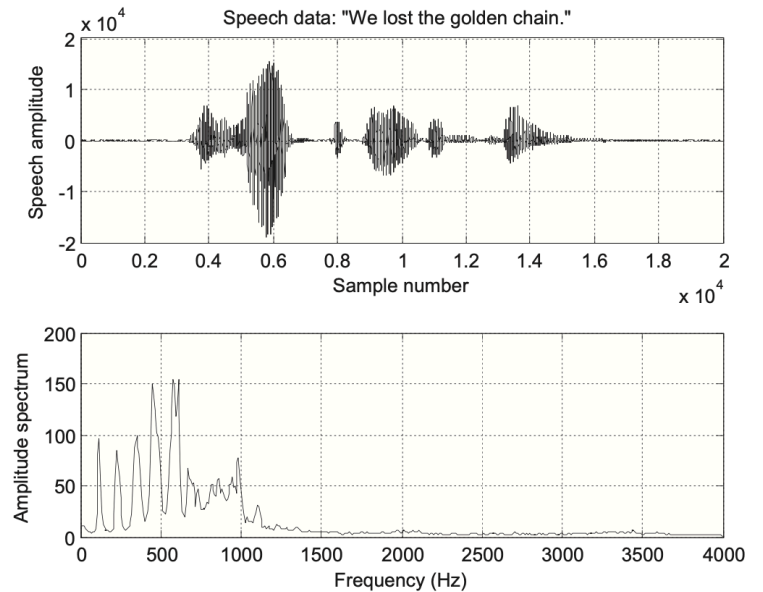


FIGURE 1.6

Speech samples and speech spectrum.

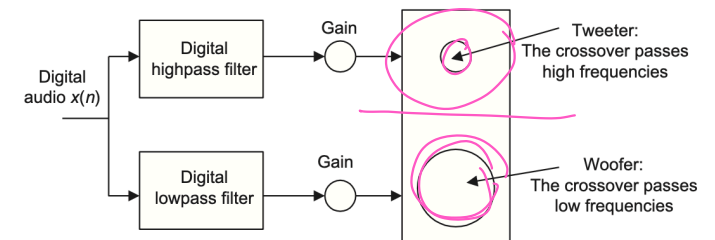
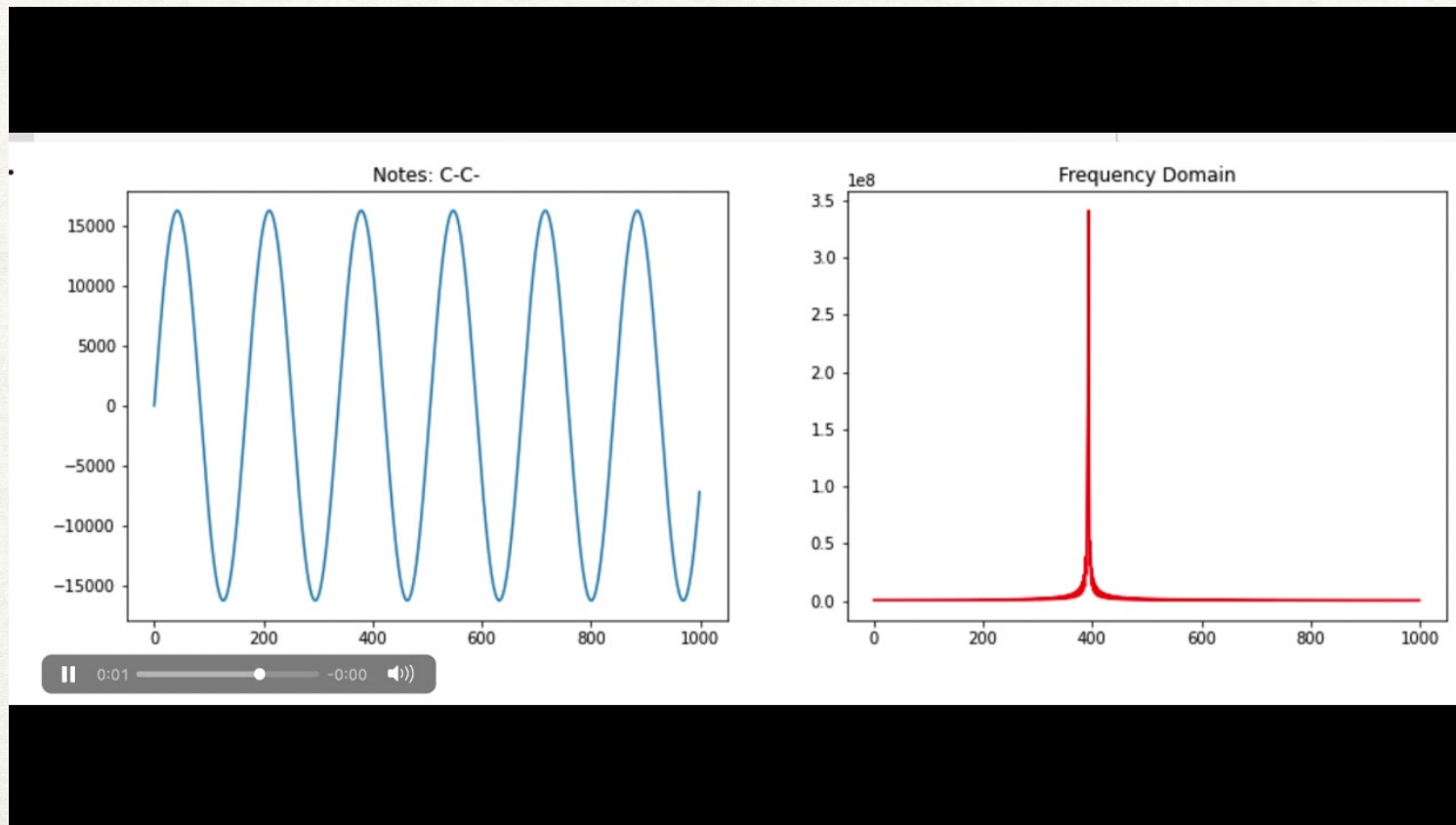


FIGURE 1.7

Two-band digital crossover.

DEMO

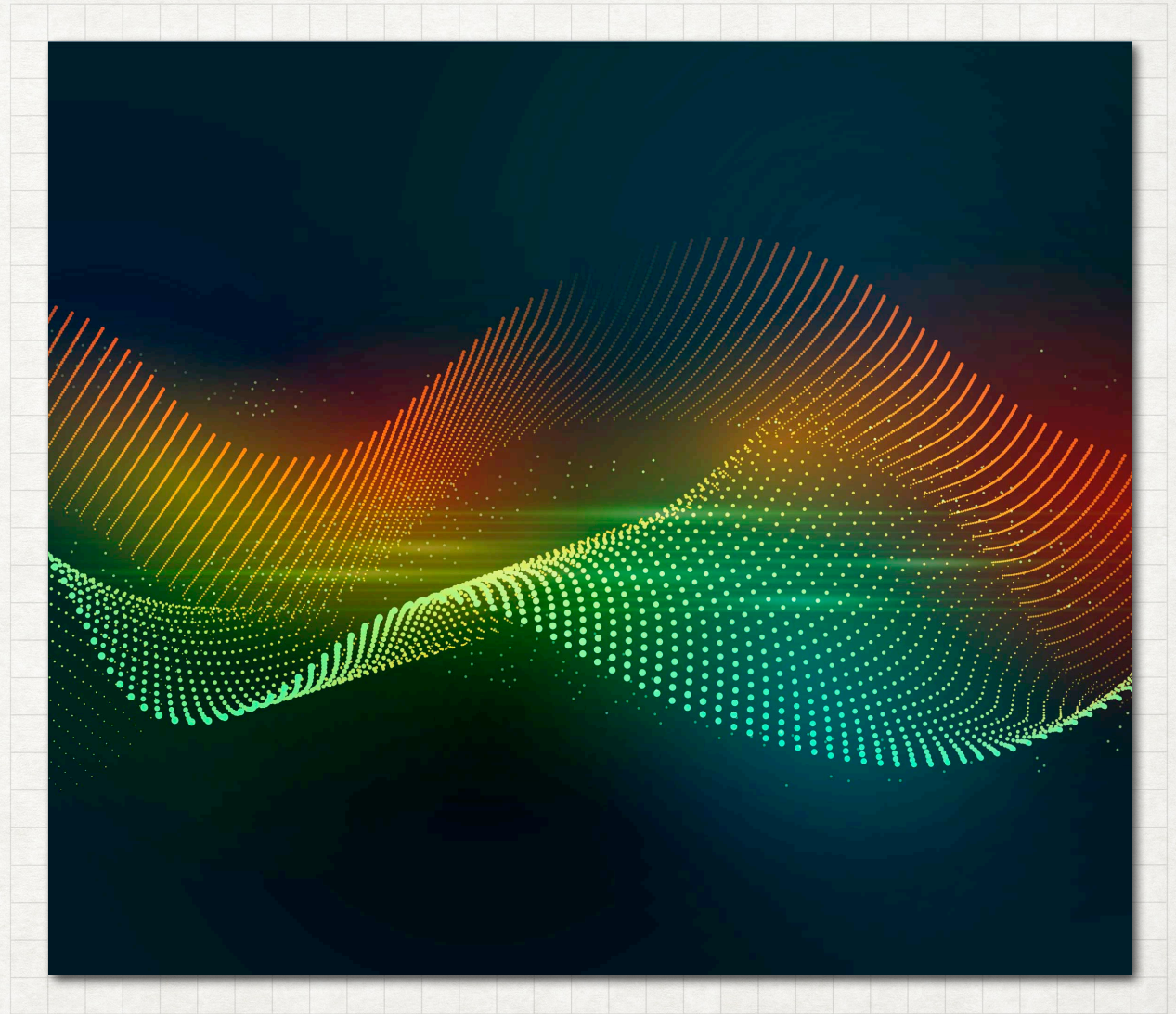
FREQUENCY DOMAIN



LECTURE 01

REPRESENTING

SIGNALS



INTRODUCTION

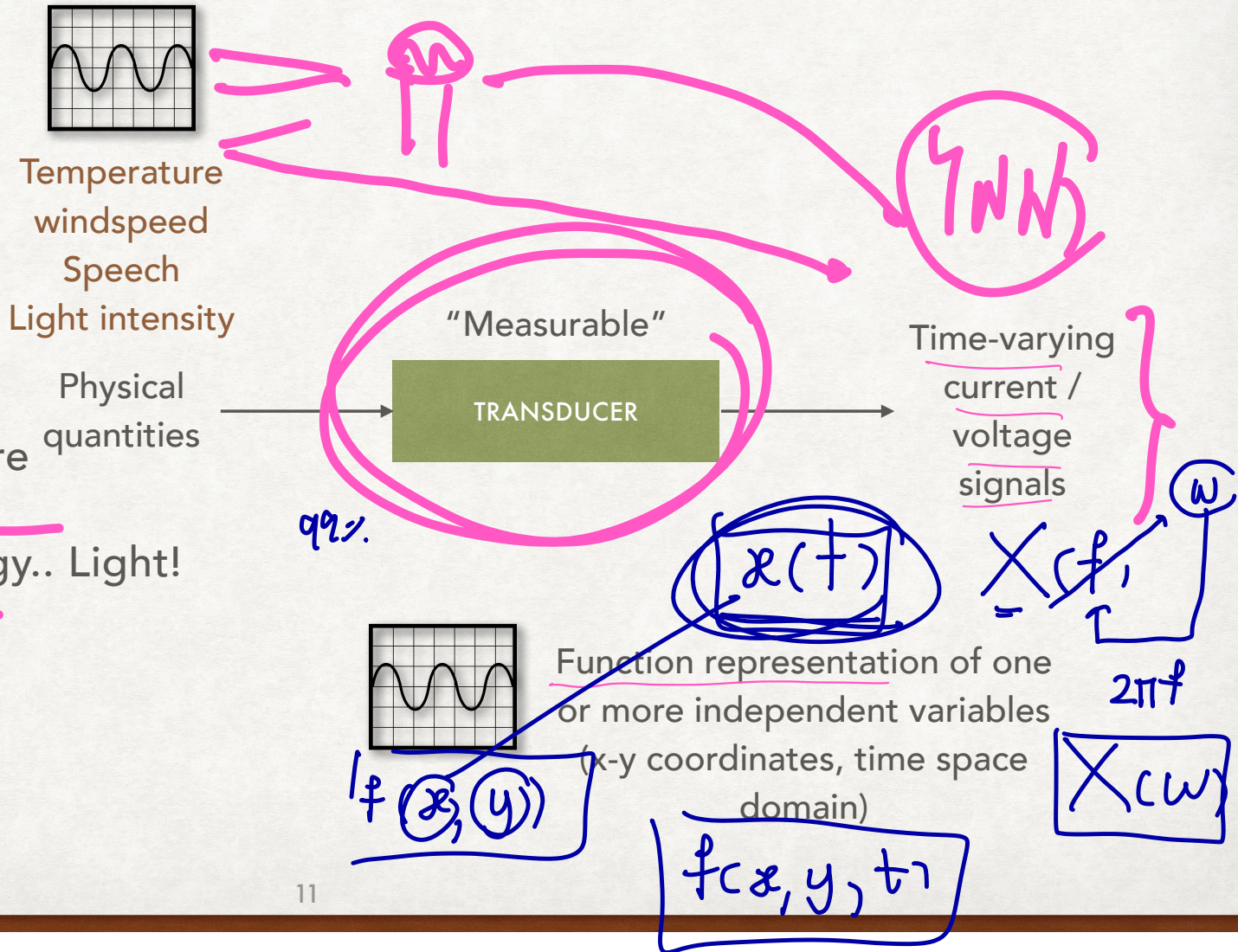
- Examples of signals?

- Human voice

- Television pictures

- Atmosphere temperature

- What we see.. EM energy.. Light!

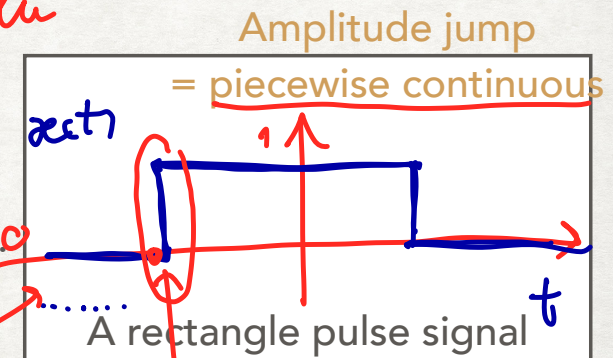
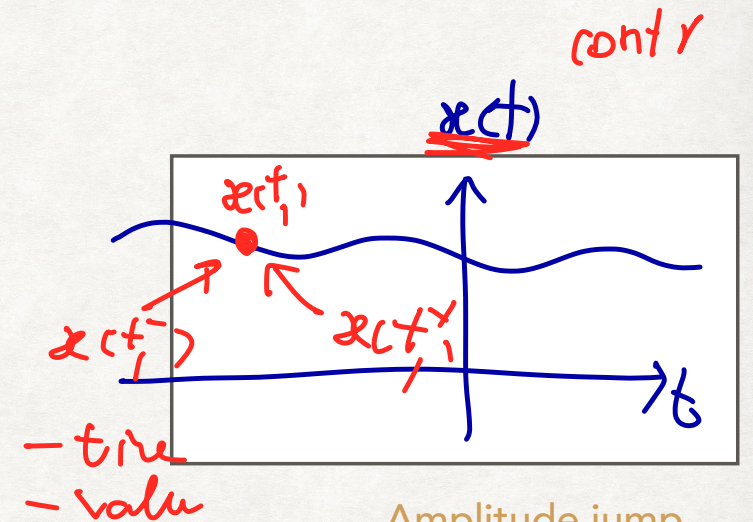


CONTINUOUS-TIME VS. DISCRETE-TIME SIGNALS

Continuous signal:

- (1) Continuous-time signal - independent variable is continuous
 - Example: telephone / audio signals or atmosphere pressure
- (2) Signal $x(t)$ is continuous if $x(t)$ is continuous of all t
 - At instant t_1 and small positive ϵ , $t_1 + \epsilon$ and $t_1 - \epsilon$ denoted as t_1^+ and t_1^- , respectively. If $x(t_1^+) = x(t_1^-) = x(t_1)$, we said $x(t)$ is continuous at $t = t_1$
 - Otherwise, it is discontinuous at t_1 and amplitude has jump at that point
- Continuous-time signal is not continuous:

$$\text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| < \frac{\tau}{2} \\ 0, & |t| > \frac{\tau}{2} \end{cases}$$



$$x(t_1^-) \neq x(t_1^+)$$

CONTINUOUS-TIME VS. DISCRETE-TIME SIGNALS

- If the independent variable takes on only discrete values

$$T_s = 12h$$

$$x(t)$$

$$x[k]$$

$$t = kT_s$$

$$0, 1, 2, 3, \dots, \underline{-1}, \dots$$

- Where T_s is a fixed positive real number and k ranges over the set of integers (i.e., $k = 0, \pm 1, \pm 2$, etc.)
- The corresponding signal $x(kT_s)$ is called a discrete-time signal
- Example - the amount of a loan payment in k th month, number of children born on a specific day - denoted as $x[k]$

CONTINUOUS-TIME SIGNALS

PERIODIC VS. APERIODIC

- Any continuous-time signal satisfies condition:

- $x(t) = x(t + nT)$ $n = 1, 2, 3, \dots$ $x(t)$ is periodic

- where $T > 0$ is a constant known as the fundamental period

- An example of a periodic signal: sinusoidal function

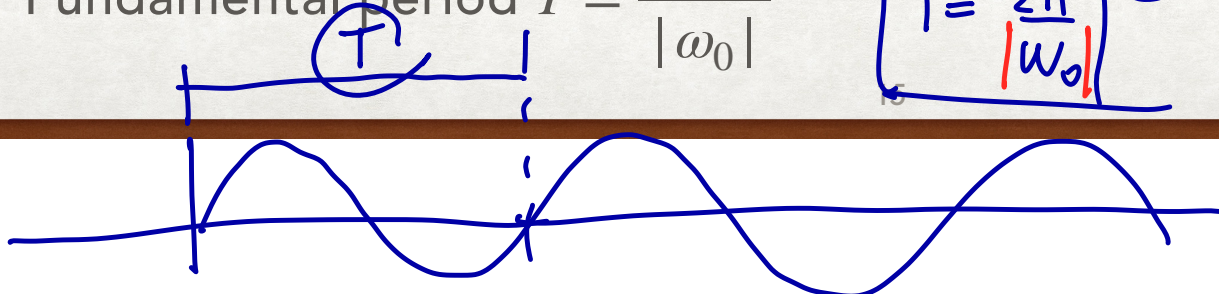
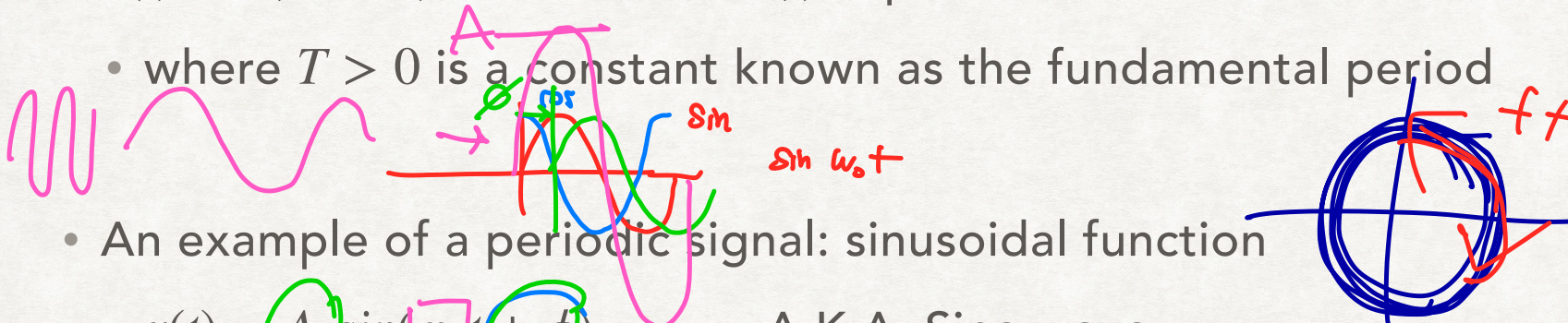
- $x(t) = A \sin(\omega_0 t + \phi)$ A.K.A. Sine wave

- Where A is an amplitude, ω_0 is a radian frequency (rad/s) and ϕ is an initial phase angle (in radian)

- Fundamental period $T = \frac{2\pi}{|\omega_0|}$

$$\omega_0 = 2\pi f = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{|\omega_0|}$$

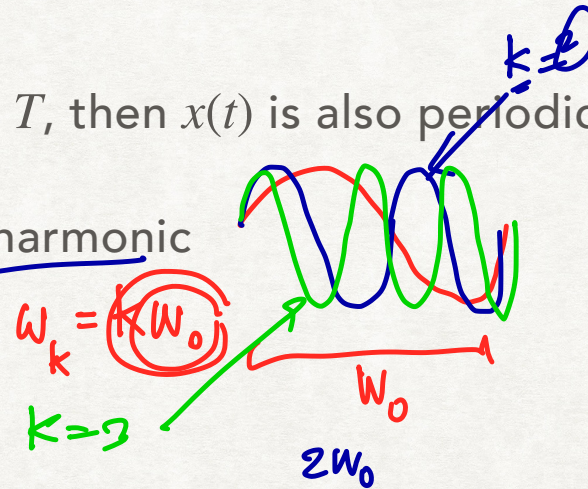


PERIODIC VS. APERIODIC

If $x(t)$ is periodic with fundamental period T , then $x(t)$ is also periodic with period $2T, 3T, \dots$

ω_0 : radian frequency $\omega_k = k\omega_0$ kth harmonic

Waveforms at each harmonic are distinct!



• $x_1(t) = \frac{1}{2} \cos 2\pi t$

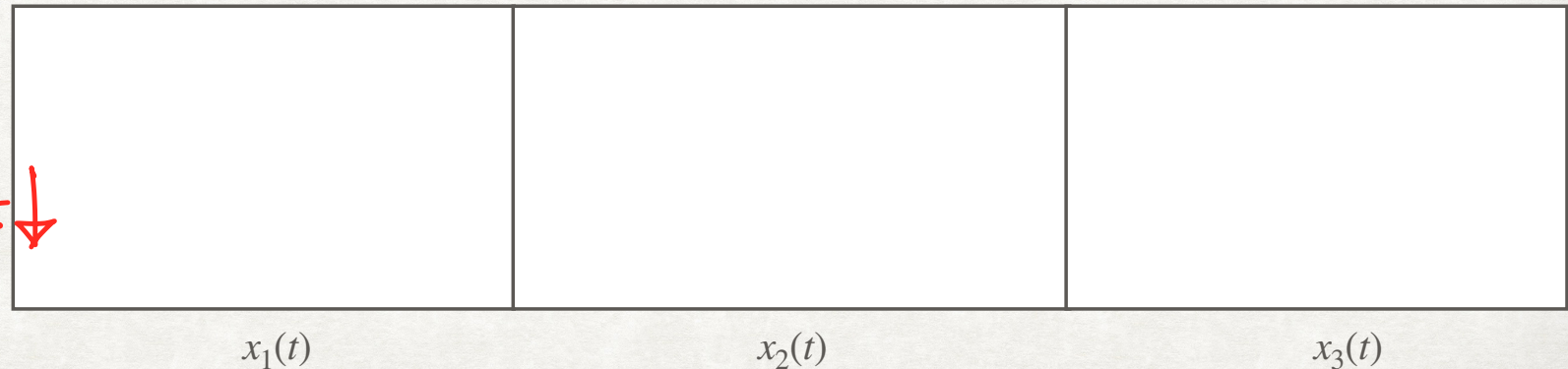
$\omega_0 = 2\pi f$ $f=1$

• $x_2(t) = \cos 4\pi t$

$f=2$ $A \uparrow$

• $x_3(t) = \frac{2}{3} \cos \pi t$

$f=0.5$ $T \uparrow$



$\sin \theta = \cos$ $\cos \theta = \sin$
 $e^A \cdot e^B = e^{A+B}$ $\cos \theta \cos \omega t$

$$z = a + j\underline{b}$$

$$e^{j\omega} \quad \boxed{e^{j\omega t}}$$

$$e^{j\alpha}$$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

PERIODIC VS. APERIODIC

EXERCISE



k^2 (mm)

Harmonically related continuous-time exponentials are sets of complex exponentials with fundamental frequencies that are all multiples of a single positive frequency ω_0 . Find the fundamental period of this signal.

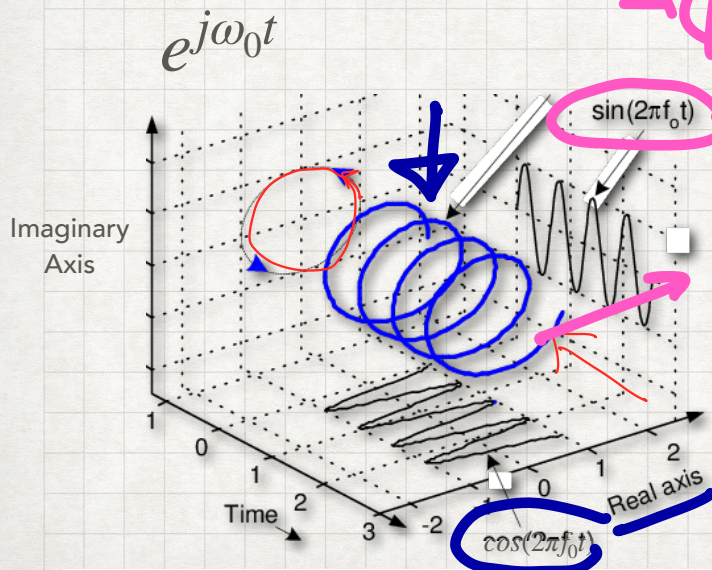
$$e^{jx} = \cos x + j \sin x$$

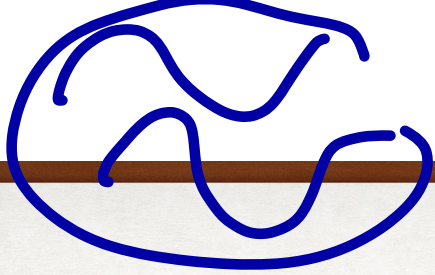
$$\phi_k(t) = e^{jk\omega_0 t}, \text{ where } k = 0, \pm 1, \pm 2, \dots$$

$$W = k(W_0)$$

$$T = \frac{2\pi}{1 \text{ kw}_0}$$

$$\frac{f}{\cancel{f}} = \frac{1}{\cancel{f}}$$





sin wt cos wt any

SP500 = sin + cos + any

PERIODIC VS. APERIODIC

- The sum of two or more periodic signals may $\neq$ periodic.
- Consider $x(t)$ and $y(t)$ with fundamental period T_1 and T_2 , respectively. Considering the sum

$z(t) = ax(t) + by(t)$ is periodic?

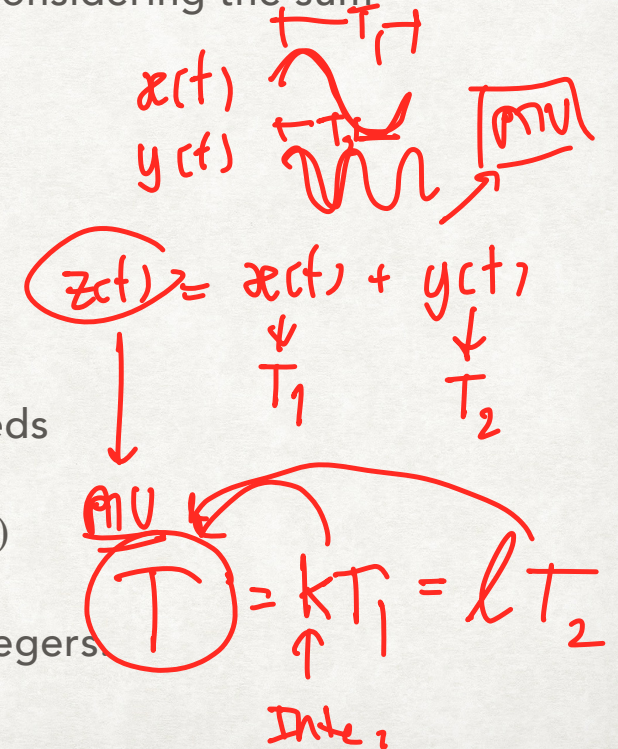
$$x(t) = x(t + kT_1), \quad y(t) = y(t + lT_2)$$

$$\text{So } z(t) = ax(t + kT_1) + by(t + lT_2)$$

In order z to be periodic with period T , one needs

$$ax(t + T) + by(t + T) = ax(t + kT_1) + by(t + lT_2)$$

We must have $T = kT_1 = lT_2$ where k and l are integers



$$x_1(t) = \sin\left(\frac{2\pi}{3}t\right) \rightarrow \boxed{T = 3 \text{ s}}$$

$\nwarrow \omega_0 \quad T = \frac{2\pi}{\omega_0}$

$$x_2(t) = \sin\left(\frac{2\pi}{5}t\right) \cos\left(\frac{4\pi}{3}t\right) \rightarrow \text{PNU?}$$

$$\sin A \cos B = \frac{1}{2} (\sin(A+B) + \sin(A-B))$$

$$x_2(t) = \frac{1}{2} \left(\sin\left(\frac{26\pi}{15}t\right) + \sin\left(\frac{14\pi}{15}t\right) \right)$$

$\nwarrow T_1 = \frac{2\pi}{26\pi/15} = \frac{15}{13}$ $\swarrow T_2 = \frac{15}{7}$

$T = kT_1 = lT_2 \quad \boxed{T = 15 \text{ s}}$

$$x_3(t) = \sin(3t) \quad T_3 = \frac{2\pi}{3}$$

$$\boxed{x_4(t) = x_2(t) + x_3(t)}$$

NU

$$T_2 = 15$$

$$\frac{2\pi}{3}$$

$$k \cdot \frac{15}{13} = l \cdot \frac{15}{7}$$

$$\frac{k}{l} = \frac{13}{7} \quad k = 13, \quad l = 7$$

$$\boxed{\frac{k}{l} = \frac{T_2}{T_1} = \frac{2\pi/3}{15}} \quad \text{+ 2x1 } k, l \text{ integer}$$

$\sim$
 $P = VI$
 $= \frac{V^2}{R}, I^2 R$
 $R=1$

ENERGY VS. POWER SIGNALS

- Signal energy over a time interval of length $2L$

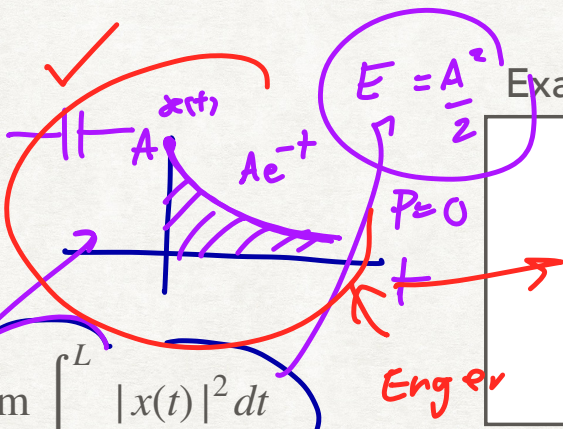
$$E_{2L} = \int_{-L}^L |x(t)|^2 dt$$

Total energy $t \in (-\infty, \infty)$: $E = \lim_{L \rightarrow \infty} \int_{-L}^L |x(t)|^2 dt$

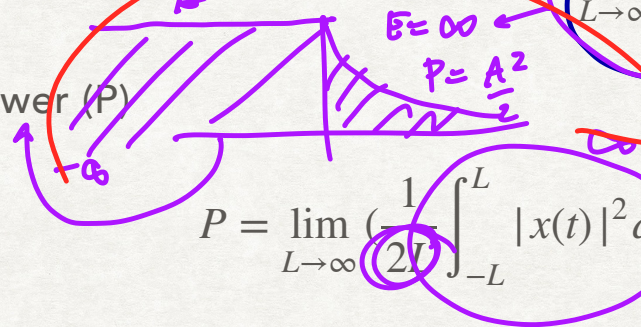
- Average power (P)

$$P = \lim_{L \rightarrow \infty} \left(\frac{1}{2L} \int_{-L}^L |x(t)|^2 dt \right)$$

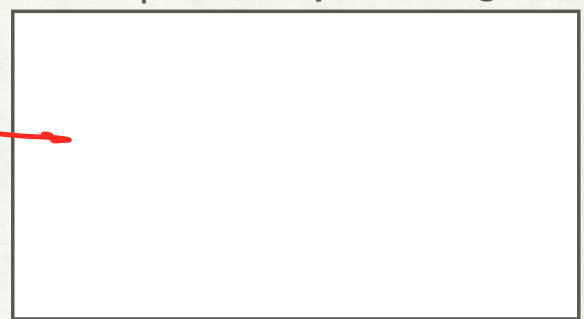
- $0 < E < \infty \rightarrow x(t)$ is an energy signal with zero power
- $0 < P < \infty \rightarrow x(t)$ is a power signal with infinite energy



Example of an energy signal



Example of a power signal

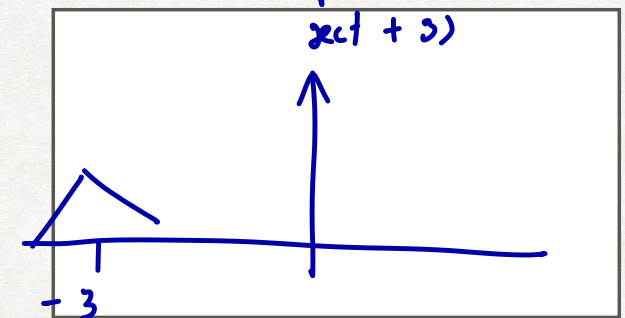
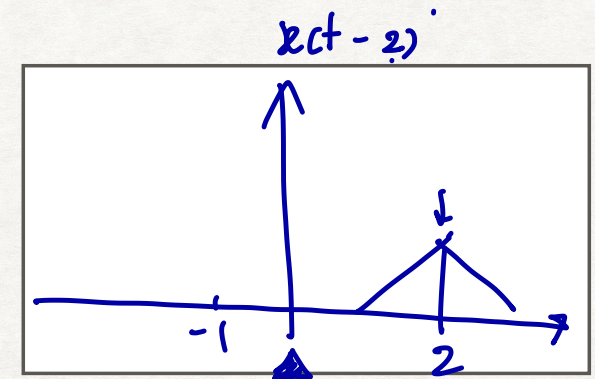
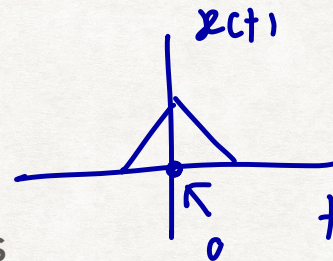


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TRANSFORMATIONS OF INDEPENDENT VARIABLE

SHIFTING OPERATION

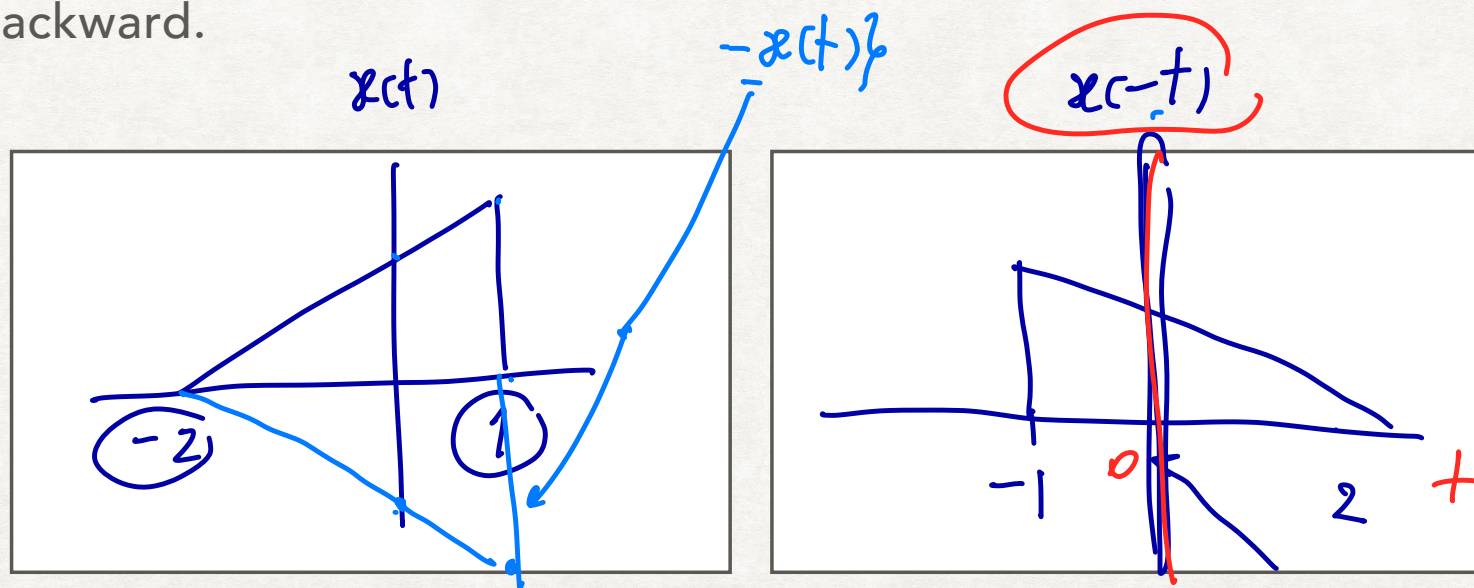
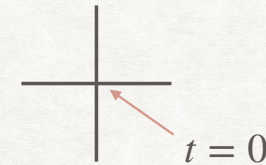
- Signal $x(t - t_0)$ is time shift version of $x(t)$
- If $t_0 > 0$: signal is delayed by t_0 seconds
- If $t_0 < 0$: signal is advanced replica of $x(t)$
- e.g., radar, sonar, communication systems, seismic



TRANSFORMATIONS OF INDEPENDENT VARIABLE

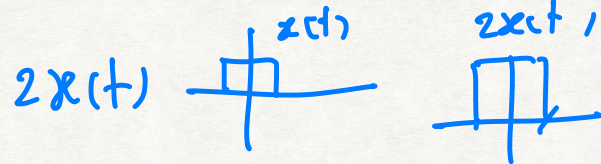
REFLECTION OPERATION

- Signal $x(-t)$: reflection of $x(t)$ about $t = 0$
- e.g., if $x(t)$: signal out of an audio recorder, then $x(-t)$: signal out of an audio player played backward.



TRANSFORMATIONS OF INDEPENDENT VARIABLE

TIME-SCALING



- Signal $x(t)$ is scaled by parameter η $x(\frac{t}{\eta})$

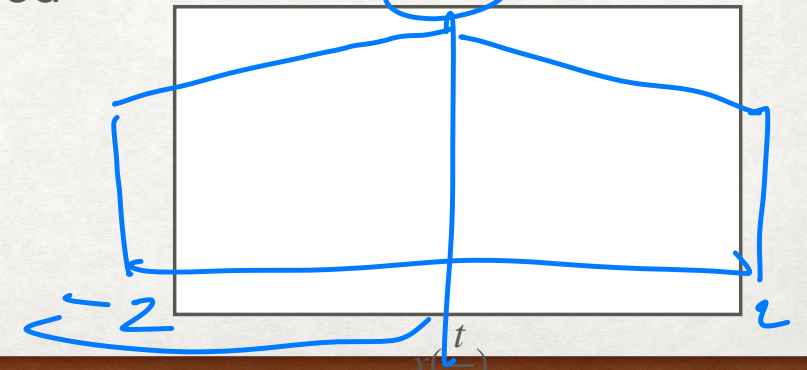
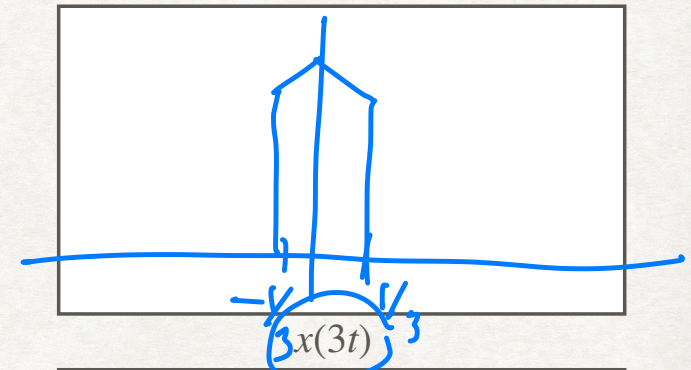
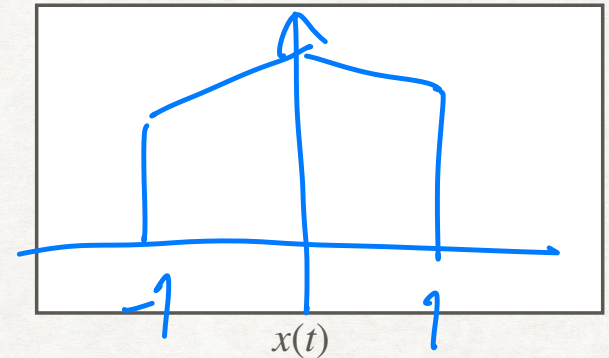
- $x(\eta t)$: a compressed version if $|\eta| > 1$

an expanded version if $|\eta| < 1$

- e.g., if $x(t)$ output of video tape recorder

$x(3t)$ -> recording is played three times speed

$x(\frac{t}{2})$ -> played at half speed



TRANSFORMATIONS OF INDEPENDENT VARIABLE



EXERCISE

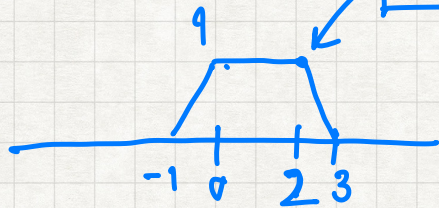
Example #2:

Plot $x(3t-6)$

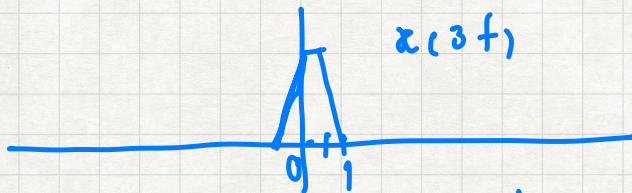
① $\text{len } 6 \rightarrow \text{scal } 3$

② $\text{scal } 3 \rightarrow \text{val } .2$

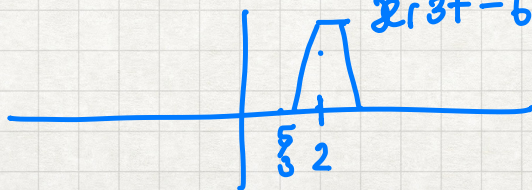
$$x(t) = \begin{cases} t+1, & -1 \leq t \leq 0 \\ 1, & 0 < t \leq 2 \\ -t+3, & 2 < t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$



$x(3t)$



$x(3t-6)$



TRANSFORMATIONS

- For any vernal signal $x(t)$, the transformation $\alpha t + \beta$ on the independent variable can be performed as follows:

$$x(\alpha t + \beta) = x(\alpha(t + \frac{\beta}{\alpha}))$$

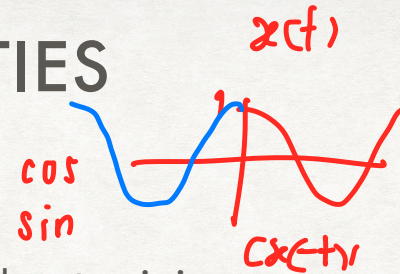
(1) Scale by α . If α is negative, reflect about the vertical axis.

(2) shift to the right by β/α if β and α have different signs, and to the left if β and α have the same sign.

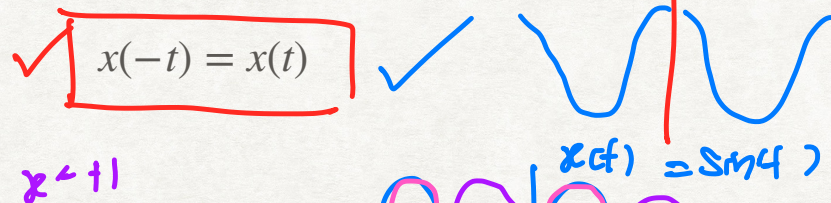
The operation of reflecting and time scaling is commutative, whereas the operation of shifting and reflecting or shifting and time scaling is not.

OTHER PROPERTIES

- Even signal - even symmetric: $x(t)$ is identical to its reflection about origin

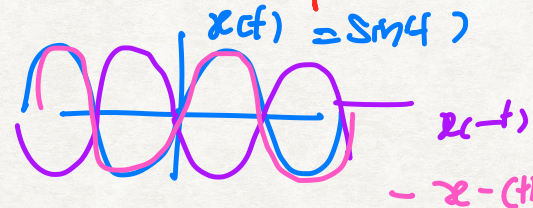


- Odd signal - odd symmetric



- Any signal $x(t)$ can be decomposed into a sum of two signals, one of which is even and one of which is odd: $x(t) = x_e(t) + x_o(t)$

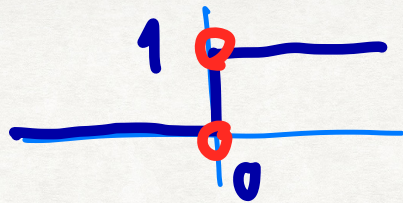
$$x(-t) = -x(t)$$



$$\text{Even part: } x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$\text{Odd part: } x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

ELEMENTARY SIGNALS



- Unit Step function (continuous-time) defined as

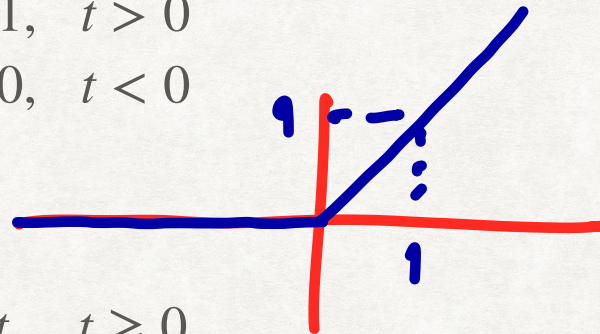
$$u(t=0) = u(0^-) + u(0^+)$$

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

- Ramp Function

$$r(t=0) = 0$$

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



Obtained by integrating unit step function

i.e., linear-sweep waveform of a cathode ray tube

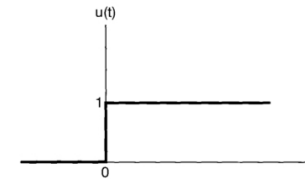


Figure 1.32 Continuous-time unit step function.

to the relationship between the discrete-time unit impulse and step functions. In particular, the continuous-time unit step is the *running integral* of the unit impulse

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau. \quad (1.71)$$

This also suggests a relationship between $\delta(t)$ and $u(t)$ analogous to the expression for $\delta[n]$ in eq. (1.65). In particular, it follows from eq. (1.71) that the continuous-time unit impulse can be thought of as the *first derivative* of the continuous-time unit step:

$$\delta(t) = \frac{du(t)}{dt}. \quad (1.72)$$

In contrast to the discrete-time case, there is some formal difficulty with this equation as a representation of the unit impulse function, since $u(t)$ is discontinuous at $t = 0$ and consequently is formally not differentiable. We can, however, interpret eq. (1.72) by considering an approximation to the unit step $u_{\Delta}(t)$, as illustrated in Figure 1.33, which rises from the value 0 to the value 1 in a short time interval of length Δ . The unit step, of course, changes values instantaneously and thus can be thought of as an idealization of $u_{\Delta}(t)$ for Δ so short that its duration doesn't matter for any practical purpose. Formally, $u(t)$ is the limit of $u_{\Delta}(t)$ as $\Delta \rightarrow 0$. Let us now consider the derivative

$$\delta_{\Delta}(t) = \frac{du_{\Delta}(t)}{dt}, \quad (1.73)$$

as shown in Figure 1.34.

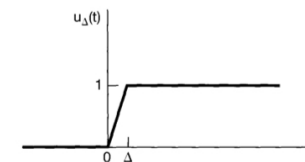


Figure 1.33 Continuous approximation to the unit step, $u_{\Delta}(t)$.

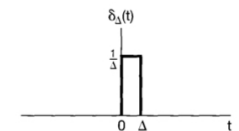


Figure 1.34 Derivative of $u_{\Delta}(t)$.

ELEMENTARY SIGNALS

- Unit Step function (continuous-time) defined as

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

- Ramp Function

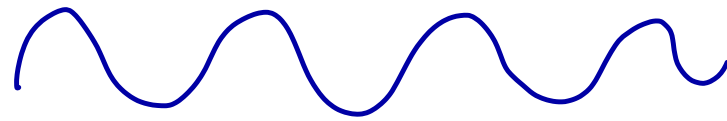
$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Obtained by integrating unit step function

i.e., linear-sweep waveform of a cathode ray tube



~~$\pi, 2\pi, 3\pi$~~



ELEMENTARY SIGNALS

$\frac{0}{0} \rightarrow \frac{\cos t}{1}$

- Sampling function

$$\text{Sinc}(x) = \frac{\sin x}{x}$$

$$\Rightarrow \text{sinc}(t) = \frac{\sin t}{t}$$

~~$Sa(x) = \frac{\sin x}{x}$ and $\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$~~

- Unit Impulse Function - or Dirac delta function or Delta function $\delta(t)$

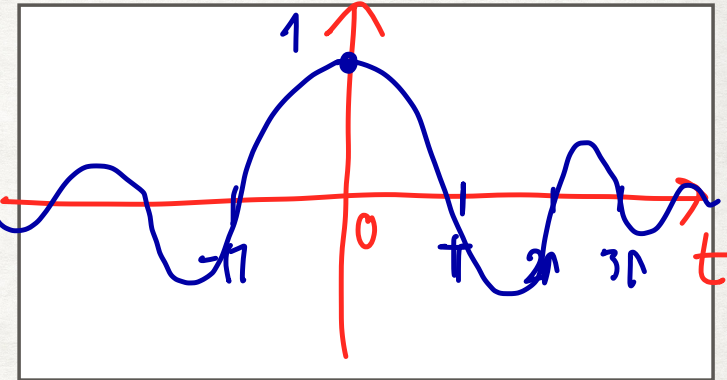
- e.g., physical phenomena such as point sources, point charges, concentrated loads on structures, voltage or current source acting very short times

$\delta(t)$

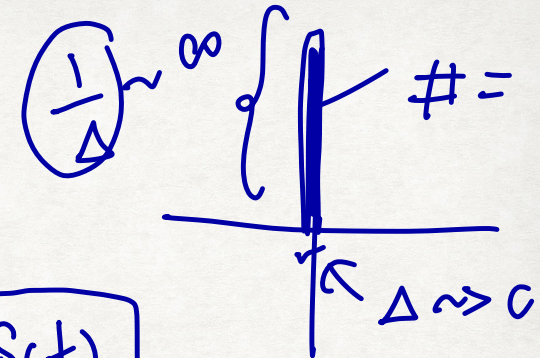
- Delta function can be defined as

$$\int_{t_1}^{t_2} x(t)\delta(t)dt = x(0) \quad : t_1 < 0 < t_2$$

- $\delta(t) \sim$ spike at origin



ELEMENTARY SIGNALS



- Unit Impulse Function - or Dirac delta function or Delta function $\delta(t)$ has the following properties

- 1) $\delta(0) \rightarrow \infty$

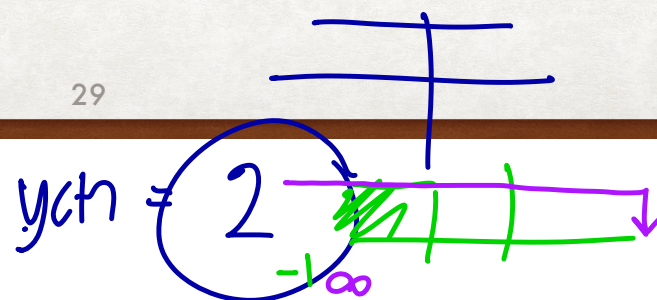
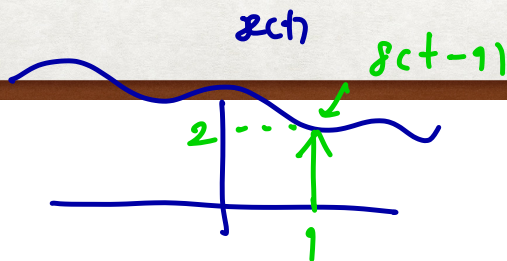
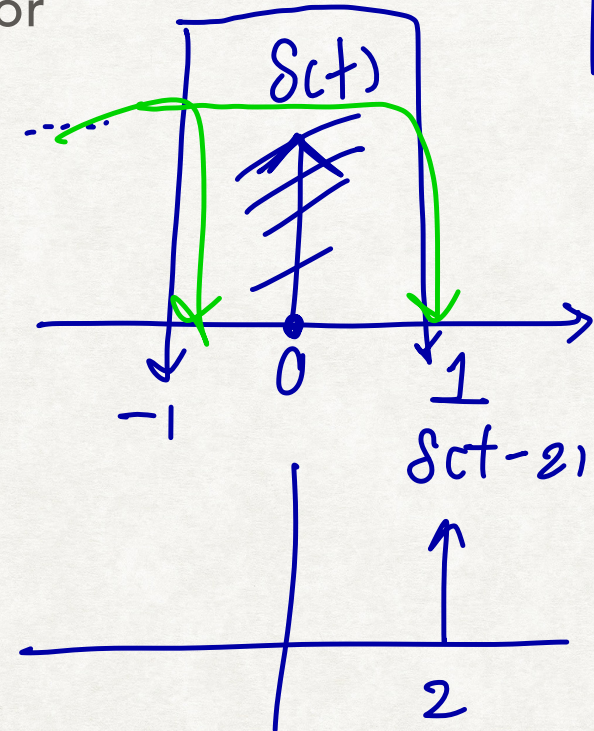
- 2) $\delta(t) = 0, \quad t \neq 0$

- 3) $\int_{-\infty}^{\infty} \delta(t) dt = 1$

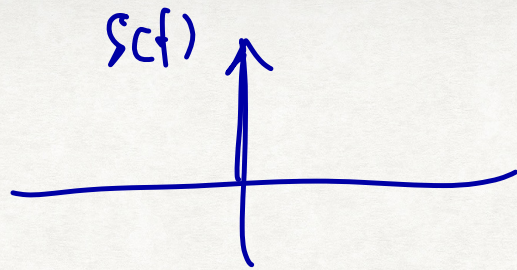
- 4) $\delta(t)$ is an even function, i.e., $\delta(t) = \delta(-t)$

$$\delta(t=0) = \infty$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



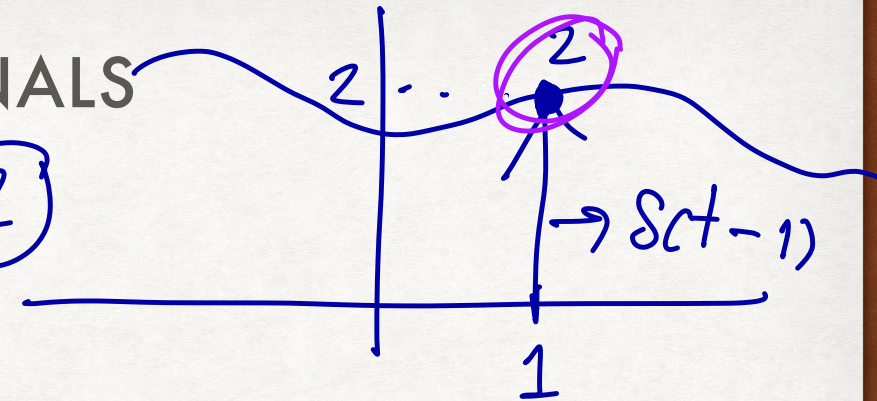
$$\int \underbrace{x(t)}_{-\infty} \delta(t-1) dt = \int_{-\infty}^{\infty} \underbrace{2}_{x(t_0)} \cdot \underbrace{\delta(t-1)}_{\delta(t_0)} dt = 0$$



Unit Impulse Function

ELEMENTARY SIGNALS

$$x(t=1) = 2$$



- When operating with delta functions, there are three additional important properties:

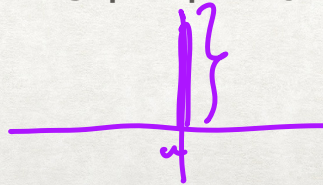
- Sifting property:

$$\int_{t_1}^{t_2} x(t) \delta(t - t_0) dt = \begin{cases} x(t_0), & t_1 < t_0 < t_2 \\ 0, & \text{otherwise} \end{cases}$$

- Sampling property:

$$x(t) \delta(t - t_0) = x(t_0) \delta(t - t_0)$$

- Scaling property:



$$\delta(at + b) = \frac{1}{|a|} \delta\left(t + \frac{b}{a}\right)$$

$$\delta(at + b) \rightarrow \delta\left(\frac{1}{a}t + \frac{b}{a}\right)$$

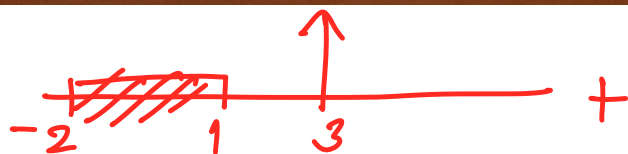
$$\frac{1}{|a|} \delta\left(t + \frac{b}{a}\right)$$

$$\begin{aligned} x(t) \cdot \delta(t-1) &= x(1) \delta(t-1) \\ &= 2 \cdot \delta(t-1) \end{aligned}$$

INTEGRAL-DERIVATIVE PAIR

$$\frac{d}{dt} u(t) = \delta(t)$$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

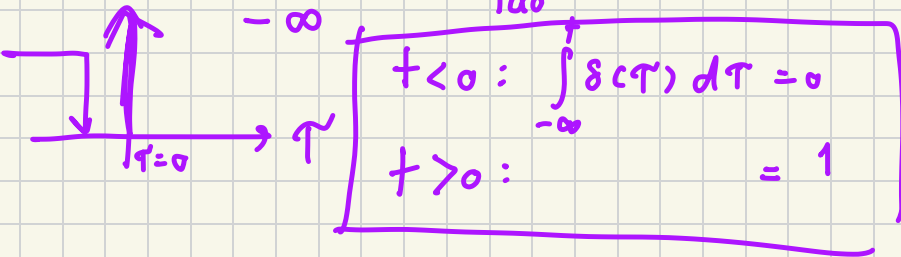


$$a) \int_{-2}^{\infty} (t+t^2) \delta(t-3) dt = \text{red circle}$$

$\xrightarrow{\text{red}} = 3+3^2 = 12$

$$b) \int_{-2}^4 (t+t^2) \delta(t-3) dt = 3+3^2 = 12$$

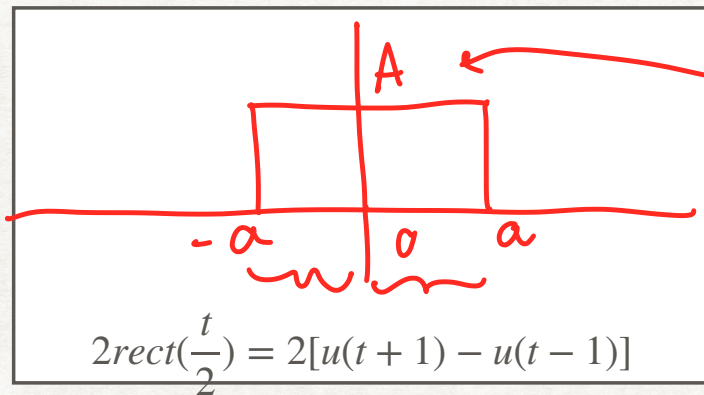
$$c) \int_{-\infty}^t \delta(\tau) d\tau = ? \quad u(t)$$



$$\frac{du(t)}{dt} = \delta(t)$$

OTHER TYPES OF SIGNALS

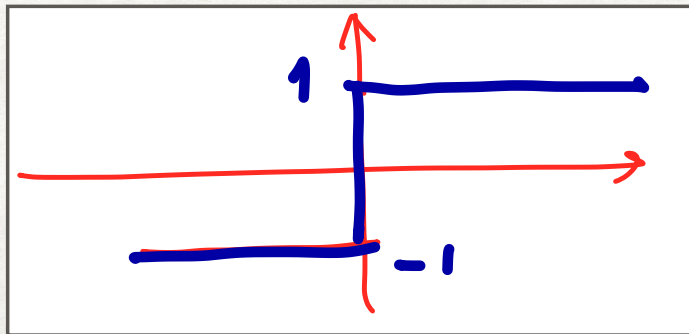
- rect : rectangular pulse signal (on-off switching)



Rectangular pulse signal extended from $-a$ to a with an amplitude A

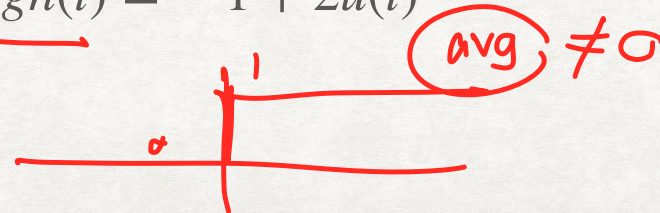
i.e., $A * \text{rect}\left(\frac{t}{2a}\right) = A(u(t+a) - u(t-a))$

- sgn (signum function)



signum

$\text{sgn}(t) = -1 + 2u(t)$



DISCRETE-TIME SIGNALS

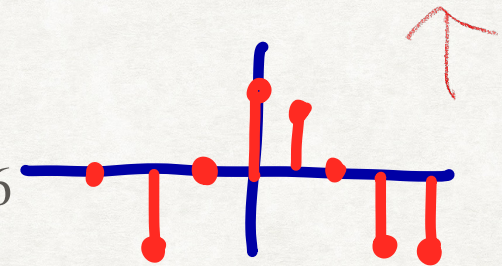
DISCRETE-TIME SIGNALS

- Signals that are defined only at discrete instants of time, i.e.,
 - number of children born on a specific day in a year
 - Number of new Covid-19 cases
- Signals occur when an analog signal is converted into a discrete-time signal by the process of sampling, i.e.,
 - Digital recording of audio signals
 - Telemetry system (data from several measurement sensors are transmitted over a single channel by time sharing)

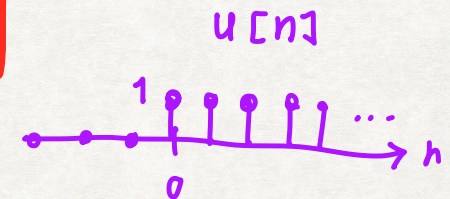
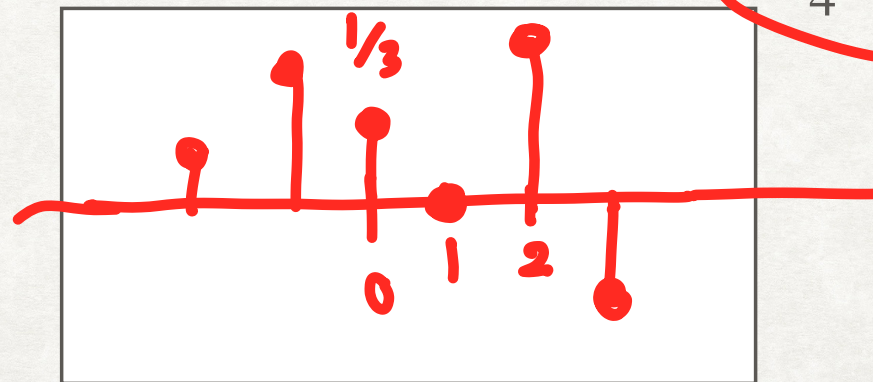
DISCRETE-TIME SIGNALS

- We represent the discrete-time signal as a sequence of values $x[n]$ with n assuming only integer values.

- Representing DT signals in functional form: $x[n] = \frac{1}{2} \cos \pi n / 6$

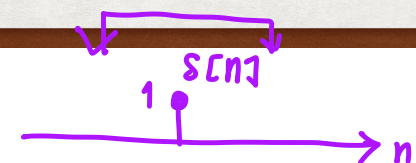


- Representing DT in as the elements of sequence: $x[n] = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}, 0, \frac{3}{4}, -\frac{1}{2}\}$



$$u[n] = 1$$

$$\sum_{n=-\infty}^{\infty} \delta[n] = 1$$



ENERGY VS. POWER SIGNALS

- Energy of a DT signal $x(n)$, for some integer $N > 0$

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

$x[n]$ IS AN ENERGY SIGNAL IF
E IS FINITE, AND
IT IS A POWER SIGNAL IF E IS
NOT FINITE, BUT P IS FINITE

- The signal $x[n]$ is periodic if, for some integer $N > 0$,

$$x[n+N] = x[n] \text{ for all } n$$

- The smallest value of N that satisfies this relation is the fundamental period of the signal
- If there is no integer N that satisfies the equation, $x[n]$ is an aperiodic signal

$$\rightarrow \sin\left(\frac{2\pi}{3}n\right) \text{ AMU?}$$

$$\rightarrow \sin(3n) \text{ AMU?}$$

$$\checkmark x[n] = x[n+N]$$

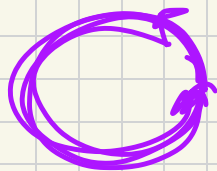
$$\cos(\omega_0 n) = \cos(\omega_0 (n+N))$$

20k $\leftarrow$ 9.4

$$\cos(\omega_0 n) = \cos(\omega_0 n + \omega_0 N)$$

AMU 75!

AMU $\rightarrow$ 9.4 100% $\checkmark$



$$\omega_0 N = \underline{\underline{2\pi \cdot k}}$$

2) N, k 100% 100% 100%

$$\cos\left(\frac{3\pi}{8}n\right)$$

$$\rightarrow \omega_0 = \frac{3\pi}{8}$$

$$\left| \frac{3\pi}{8}N = 2\pi \cdot k \right|$$

k, N

$$N = 16$$

$$\frac{k}{N} = \frac{3}{16}$$

$$k = 3$$

$$N = 16$$

$\star$

$$\cos(0.2n)$$

$$\omega_0 = 0.2$$

$$0.2 \cdot N = \underline{\underline{2\pi \cdot k}}$$

$k, N \rightarrow$ 100%

$\rightarrow$ 100% 100%

① Discrete AMU? $\rightarrow$ 11

② $+ - \checkmark$ AMU

AMU

Cont $\rightarrow$ ① Sin/cos $\checkmark \checkmark$ AMU

② 100% $\rightarrow$ 100%

$$\frac{k}{N} = \frac{T}{T_c}$$

TRANSFORMATIONS OF INDEPENDENT VARIABLE

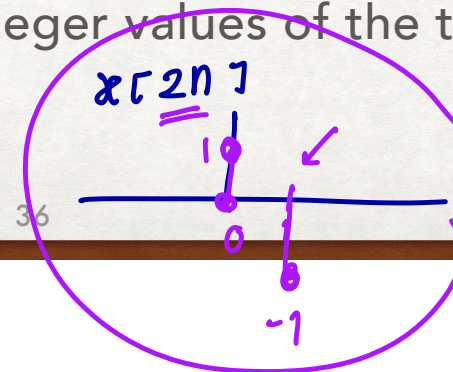
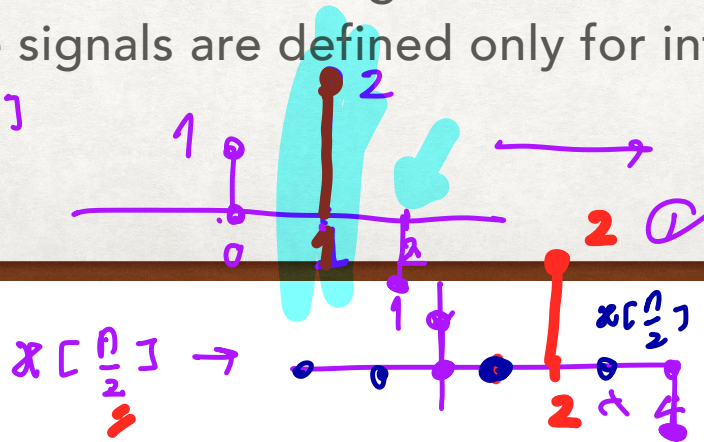
- For integer value of k , sequence $x[n-k]$ represents $x[n]$
- Shifted by k samples
 - Shift to the right if $k > 0$
 - Shift to the left if $k < 0$
- $x[-n]$ corresponds to reflecting $x[n]$ at around origin $n = 0$
- **Scaling** is similar to CT signal, but must be interpreted with care in discrete-time case, since the signals are defined only for integer values of the time variable.

$$x[n-2]$$

$$n+3$$

$$x\left[\alpha\left[n+\frac{\beta}{\alpha}\right]\right]$$

$$x[n]$$



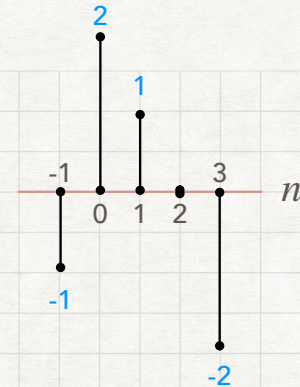
TRANSFORMATIONS OF INDEPENDENT VARIABLE

EXERCISE



Consider the waveform shown in the figure on the right, and let

$$y[n] = x\left[-\frac{n}{3} + \frac{2}{3}\right]$$



ELEMENTARY SIGNALS



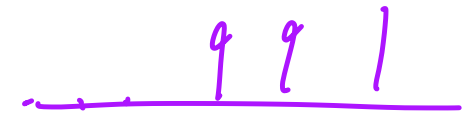
- **Unit Impulse Function** in discrete time is defined as

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

- $\delta[n]$ is somewhat similar to the continuous-time impulse function $\delta(t)$ we note that the magnitude of the discrete impulse is always finite.

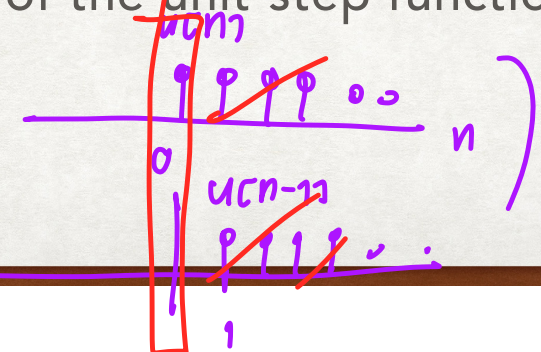
- **Unit-step sequence** is defined as

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



- The discrete-time delta and step functions have properties somewhat similar to their continuous-time counterparts. The first difference of the unit-step function is

$$u[n] - u[n-1] = \delta[n]$$



ELEMENTARY SIGNALS

Other properties

$$\sum_{k=-\infty}^n \delta[k] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases} = u[n]$$

$$\sum_{k=0}^{\infty} \delta[n-k] = u[n]$$

Similar properties to the continuous-time system:

$$x[n]\delta[n-k] = x[k]\delta[n-k]$$

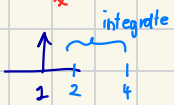
General form:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

REFERENCES

- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)
 - Chapter 1 and Chapter 6 (6.1-6.2)
- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
 - Chapter 1 (1.1-1.4)

1) $\omega_0 = 4\pi$
 $T = \frac{2\pi}{4\pi} = 0.5$ ✗

2) $t_0 = 1$ 

$t^2 + 2t + 1 = 0$ ✗

3) $T_1 = \frac{2\pi}{10}$ $T_2 = \frac{2\pi}{4}$

$T = kT_1 = lT_2$

$\frac{k}{l} = \frac{2\pi/4}{2\pi/10} = \frac{5}{2}$

$T = 5 \cdot \frac{2\pi}{10} = 2 \cdot \frac{2\pi}{4} = \pi$ ✗

